

Luigi Casagrande

Junior Manufacturing Systems Engineer

luigicasagrande.it [in luigi-casagrande](https://in.luigi-casagrande)

Experience

Production Engineer Intern > **ESSILORLUXOTTICA**

April 2026 – Current

 Sedico, BL

- Analysed MES production data to monitor performance, quality, cycle time and scrap trends, supporting root cause analysis and improvement actions.
- Contributed to Lean Manufacturing and continuous improvement initiatives to enhance workflows, resource allocation and productivity
- Developed data and software solutions to standardise departmental processes, resolve operational issues and improve efficiency

Education

MS POLITECNICO DI MILANO, **Materials Engineering and Nanotechnology** (English)

Sep 2023 – Jul 2026

Milano, Italy 

- **Master's Thesis:** Solid-state recycling of aluminium machining chips through cold compaction and hot rolling, with analysis of process parameters, microstructure and mechanical properties.
- ★ Relevant coursework: “Metallic Materials and Product Innovation”, “Composite Materials”, “Life Cycle Assessment”, “Materials for Energy”

EX SHANGHAI JIAO TONG UNIVERSITY, **Materials Engineering**

Sep 2024 – Feb 2025

Shanghai, China 

- *Worldwide Exchange Program* offered by Politecnico di Milano
- ★ Projects: “Physical aging in glassy polymers: stress relaxation and creep compliance analysis”, “Molecular dynamics modelling of melting behaviour in pure Ni, Zr, and Ni-Zr alloys”

BS POLITECNICO DI MILANO, **Materials Engineering and Nanotechnology** (Italian)

Sep 2020 – Jul 2023

Milano, Italy 

- **Bachelor's Thesis:** Printed Devices for Energy Storage
- ★ Relevant coursework: “Steelmaking”, “CAD Modelling”, “Data Analysis and Experimental Methods”, “Structural Mechanics”

EX KAWANA WATERS HIGH SCHOOL, **STEM subjects**

Jul 2016 – Sep 2016

Queensland, Australia 

- Independent study term abroad during High School

Skills

Languages: • Italian (*native*) • English (*fluent*) • Chinese (*elementary*) • French (*elementary*)

Manufacturing & Data: SAP · MES · Microsoft Office · Power BI · JMP Pro

Programming: Python · LaTeX · JavaScript · C++ · HTML/CSS · SQL

Engineering Software: Autodesk Inventor · SolidWorks · Abaqus FEA · COMSOL · MATLAB

Scientific & Other Computing Environments: GNU/Linux · openLCA · LAMMPS

Certificates

IELTS Academic < issued by: *British Council*

English language certification, C1 Level (7.5)

2023

International Pool Lifeguard Certification < issued by: *Federazione Italiana Nuoto*

2017

Pre-Academic Piano Certificates < issued by: *Conservatori di Musica italiani*

2013 & 2016